

Keshav Kacholiya

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WORK EXPERIENCE

Founding Engineer – Pentesting & AI Automation — Fenrir Security

Aug 2025 – Present

- Conducted **web application pentests** on real client targets and practice apps (demo.testfire.net, DVWA, HackTheBox) — documented **104 structured vulnerability reports** mapped to **XBOW benchmarks**, covering SQLi, XSS, IDOR, and auth bypass.
- Built and integrated an **agentic AI pipeline** that automates recon, scan orchestration, and report generation — reducing manual pentesting overhead significantly.
- Analyzed **HTTP traffic, headers, and server logs** to trace root causes; validated fixes and iterated on detection logic for recurring vulnerability patterns.

Cybersecurity Intern — Network Intelligence India

Jul 2024 – Dec 2024

- Monitored system logs and correlated security events across multiple sources to surface anomalies, suspicious login patterns, and potential intrusion indicators in real-time.
- Supported **root cause analysis** of security incidents by mapping application workflows, system behavior, and log traces — contributed to closing **detection coverage gaps** in monitoring pipelines.
- Collaborated with the security team to refine alerting rules, reduce false positives, and document findings in structured incident reports for client review.

PROJECTS

AI-Powered Penetration Testing Agent

- Built an **agentic AI system** that autonomously runs recon, vulnerability scanning, and structured report generation using **LLM-driven tool-use workflows**.
- Integrated **Nmap, Burp Suite, and custom Python modules** with an AI agent; outputs severity-rated findings with CVE references and remediation steps.
- Tech Stack:** Python, LLM APIs, Agentic AI Framework, Nmap, Burp Suite, Bash

CTF Lab Environment (Docker + Python)

- Designed and deployed **isolated CTF challenge labs** using **Docker** — each container simulates a real vulnerability category (web, privilege escalation, reverse engineering).
- Wrote **Python automation scripts** for lab spin-up, flag validation, and challenge scoring — enabling repeatable, self-hosted security training environments.
- Tech Stack:** Docker, Python, Linux, Bash, Networking

SIEM Detection Lab & Security Automation Toolkit

- Built a **SOC simulation lab** with detection use cases for brute-force, lateral movement, and abnormal login patterns — iterated on alert tuning to reduce false positives.
- Developed Python tools including an **HTTP header analyzer, subdomain enumerator, and log anomaly detector** to automate repetitive security analysis tasks.
- Tech Stack:** Python, Bash, Linux, SIEM Concepts, Log Analysis

CERTIFICATIONS & RECOGNITION

Cisco – Ethical Hacker Cisco Networking Academy

Covers ethical hacking methodology, reconnaissance, exploitation techniques, and responsible disclosure practices.

Cisco – Linux Essentials Cisco Networking Academy

Foundational Linux command-line, file system management, scripting, and system administration concepts.

HackTheBox – Top 10% Global Ranking

Ranked in the **top 10% globally** — solved machines across web exploitation, privilege escalation, and Active Directory attack paths.

EDUCATION

2022 – Present **B.E. in Electronics & Communication Engineering at Atharva College of Engineering, Mumbai University** CGPA: 8.1
Class 12 (RBSE): **96.8%** | Class 10 (RBSE): **86%**

SKILLS

Cybersecurity: Web App Pentesting, OWASP Top 10, Vulnerability Research, XBOW Benchmarking, CTF
AI / Automation: Agentic AI, LLM APIs, Prompt Engineering, AI-driven Security Pipelines, Python Automation
Security Analysis: Log Analysis, SIEM, Detection Engineering, Root Cause Analysis, Threat Triage
Tools: Burp Suite, Nmap, Wireshark, Postman, Docker, Metasploit (Basic)
Programming: Python, Bash **Systems:** Linux, Windows, Git, GitHub